EE121 Discussion 3

Julian Maravilla 9/25/25

Agenda

- Angle Modulation
  - Phase Modulation (PM), Frequency Modulation (FM)
- Bandwidth Estimation
- I and Q of an FM Signal

Angle Modulated Signal

$$u(t) = 20\cos(2000\pi t + \phi(t))$$

What's the approximate bandwidth?

$$\boxed{\Delta f = \frac{1}{2\pi}\frac{d\phi}{dt}}$$

Frequency Deviation

$$\boxed{B_{FM} \approx 2B + 2\Delta f_{max} = 2B(\beta + 1)}$$

Carson's rule

Angle Modulated Signal

$$u(t) = 20\cos(2000\pi t + \phi(t))$$

What's the approximate bandwidth?

$$\phi(t) = 0.1m(t) \qquad m(t) = \cos(2\pi t)$$

$B_{FM} = 2B + 2\Delta f$, $B = 1\,Hz$, $\Delta f = \frac{1}{2\pi}\frac{d\phi}{dt} = -0.1\sin(2\pi t)$

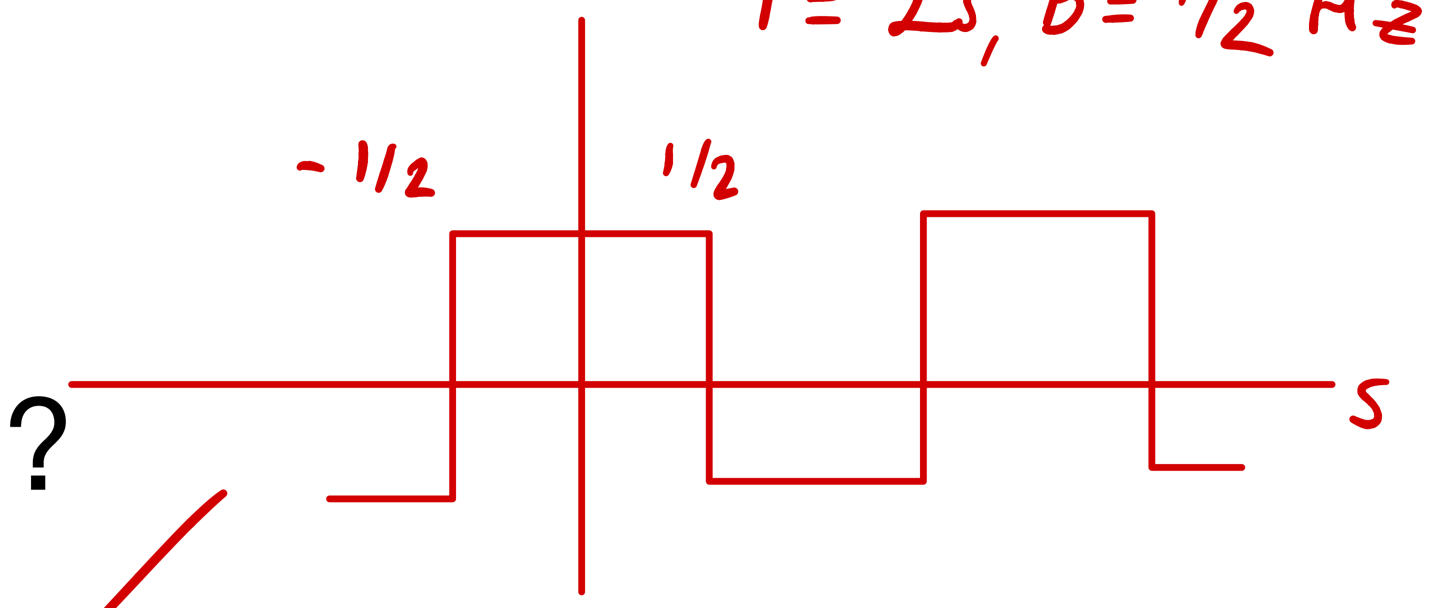
$\Delta f_{max} = 0.1\,Hz$

$$\boxed{B_{FM} = 2.2\,Hz}$$

Angle Modulated Signal

T= 2s, B= 1/2 Hz

$$u(t) = 20\cos(2000\pi t + \phi(t))$$



What's the approximate bandwidth?

$$\phi(t) = 20\pi \int_{-\infty}^{t} m(t)\, d\tau + a \qquad m(t) = \sum_{n=-\infty}^{\infty} (-1)^n I_{[-1/2,1/2]}(t-n)$$

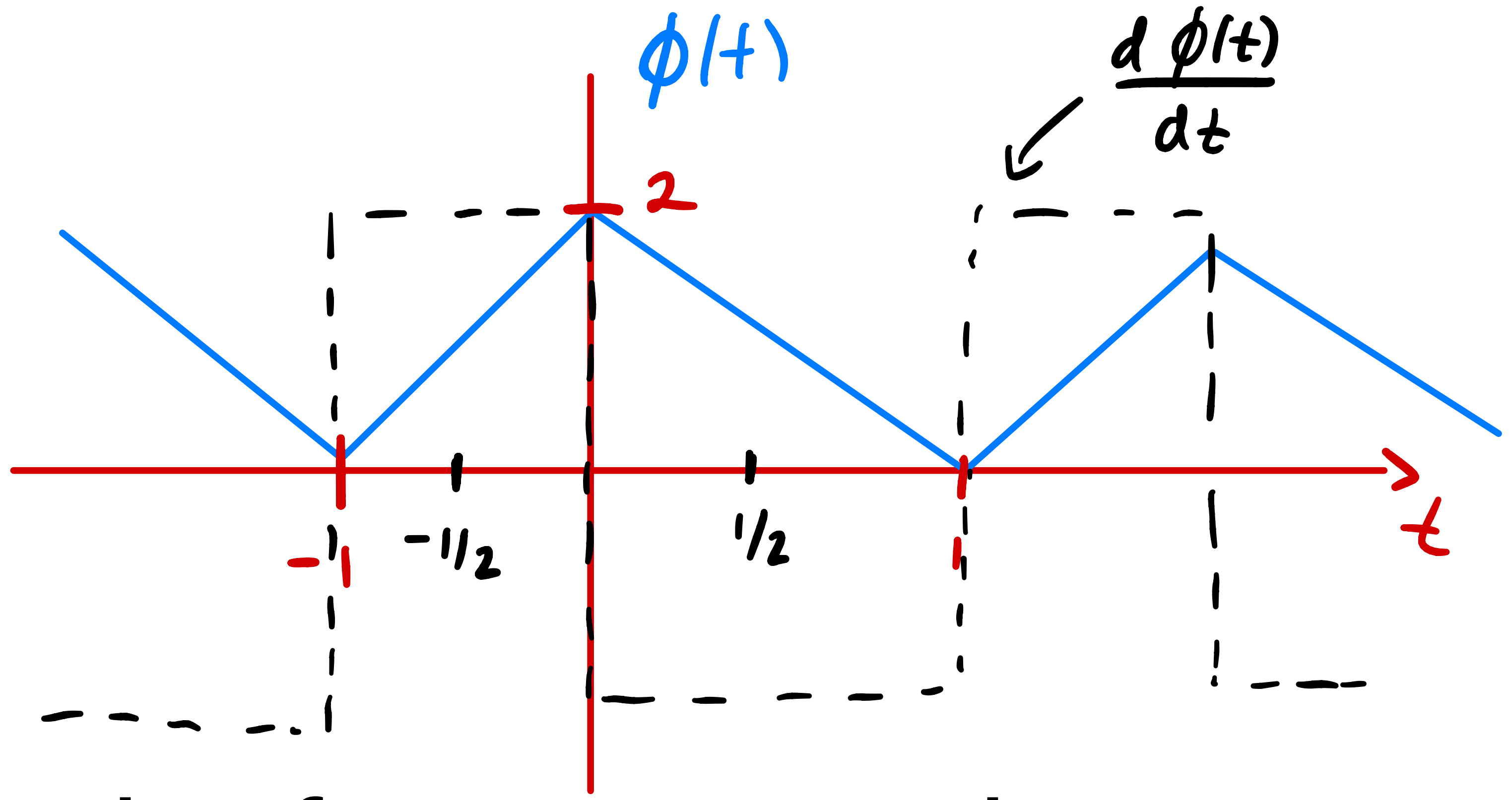
$$\phi(t) = 2\pi k_f \int_{-\infty}^{t} m(\tau)\, d\tau + a$$

$$B_{FM} = 2B + 2\Delta f_{max}$$

$$\boxed{B_{FM} = 21\text{Hz}}$$

$f(t) = k_f m(t)$, $k_f = 10$, $\Delta f_{max} = 10\text{Hz}$, $B = 1/2$ Hz

Angle Modulated Signal



$$u(t) = 20\cos(2000\pi t + \phi(t))$$

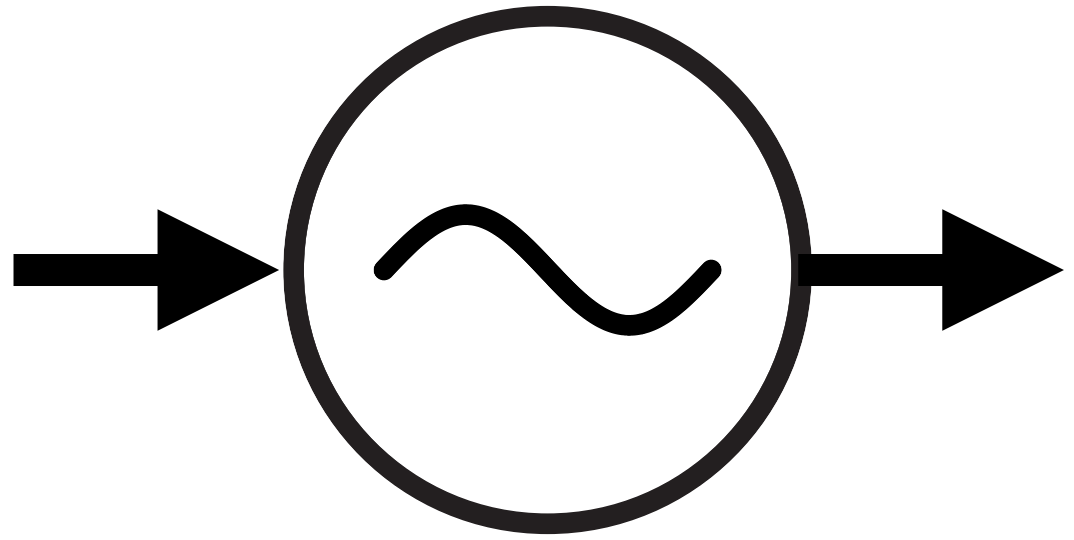
What's the instantaneous frequency deviation from the carrier frequency when:

$$\phi(t) = 2\sum_{n=-\infty}^{\infty} s(t-2n) \qquad s(t) = (1-|t|)I_{[-1,1]}(t)$$

$$\boxed{f(t) = \frac{1}{2\pi}\frac{d\phi(t)}{dt} = \frac{1}{\pi}\sum_{n=-\infty}^{\infty}\frac{ds(t)}{dt} = \frac{1}{\pi}\sum_{n=-\infty}^{\infty}(-1)^{n+1} I_{[-1/2,1/2]}(t+1/2-n)}$$

Voltage Controlled Oscillator (VCO)

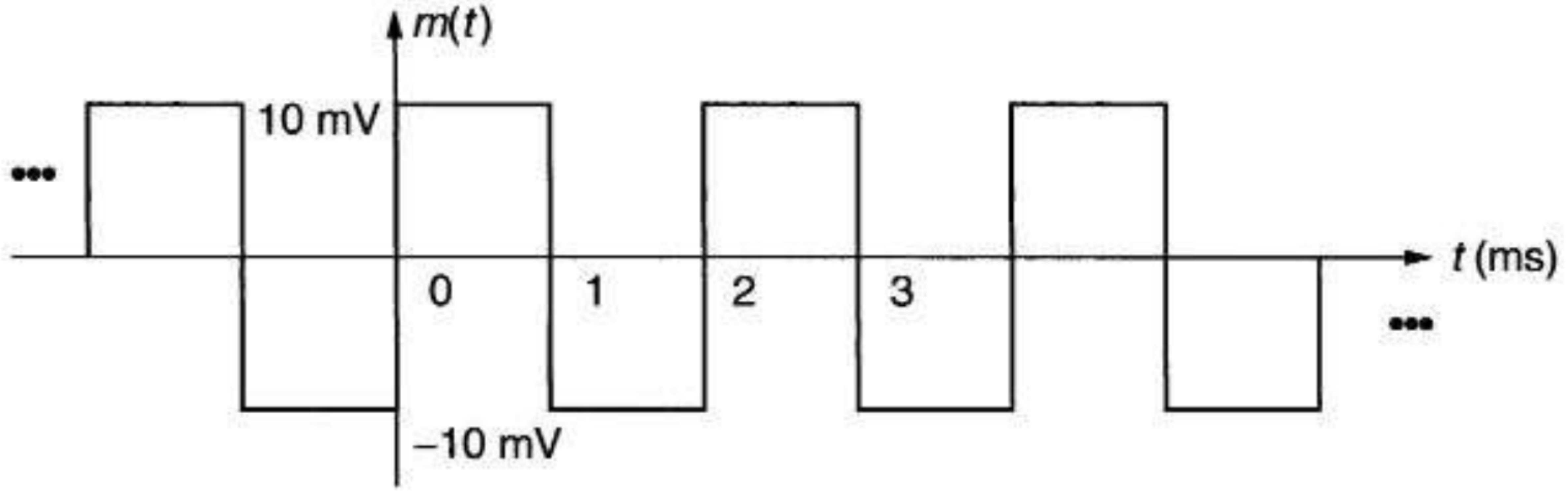
$m(t)$



$f = f_c + K_v m(t)$

Quiescent Frequency: f_c

Frequency Deviation: K_v



$f_c = 200$ KHz

$K_v = 1$ kHz/mV

What's the frequency range when the VCO is driven by $m(t)$?

$$f = f_c + K_v m(t) = 200\text{KHz} \pm \frac{1\text{KHz}}{\text{mV}} \cdot 10\text{mV} = 200\text{KHz} \pm 10\text{KHz}$$

I and Q of an FM Signal

$$I = A\cos(2\pi \cdot 10\text{kHz} \cdot t)$$

$$Q = \begin{cases} A\sin(2\pi \cdot 10\text{kHz} \cdot t), & \text{when } m(t) > 0 \\ -A\sin(2\pi \cdot 10\text{kHz} \cdot t), & \text{when } m(t) < 0 \end{cases}$$

What's the output of the VCO?

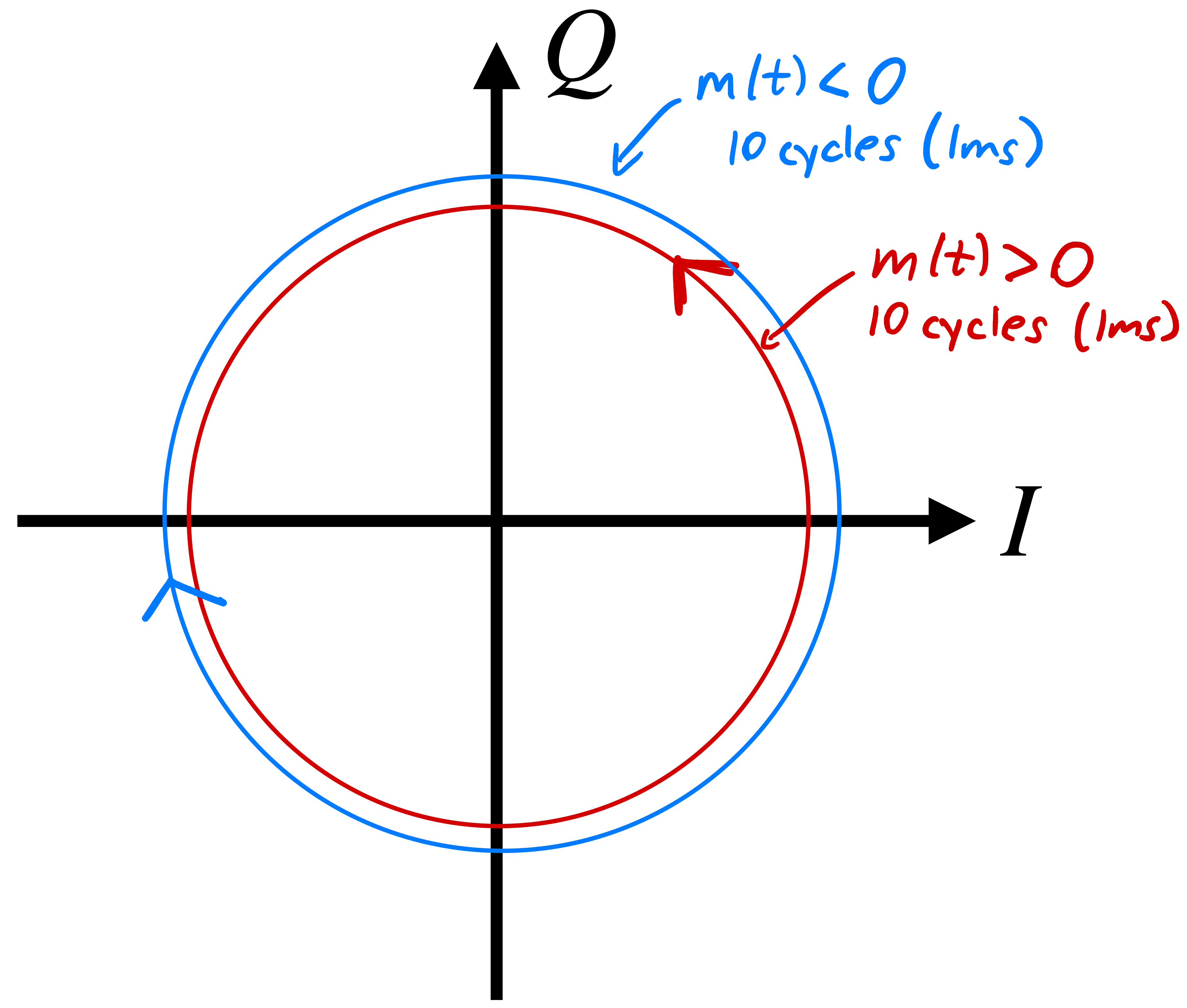
$$u(t) = A\cos(2\pi f(t) t)$$

$$= A\cos(2\pi f_c t + 2\pi k_v m(t) t)$$

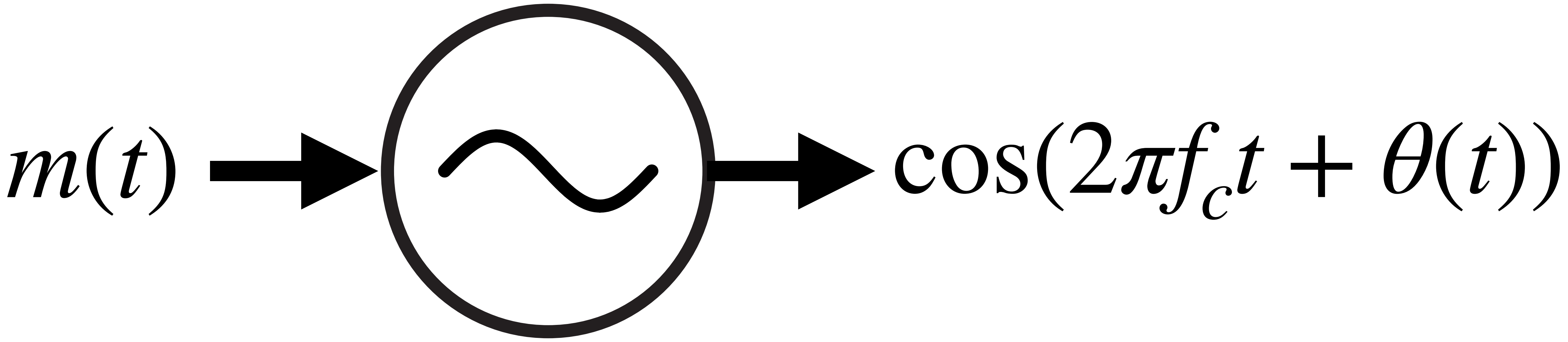
$$= \mathrm{Re}\left\{A e^{j2\pi k_v m(t) t} e^{j2\pi f_c t}\right\}$$

$$I = A\cos(2\pi k_v m(t) t)$$

$$Q = A\sin(2\pi k_v m(t) t)$$

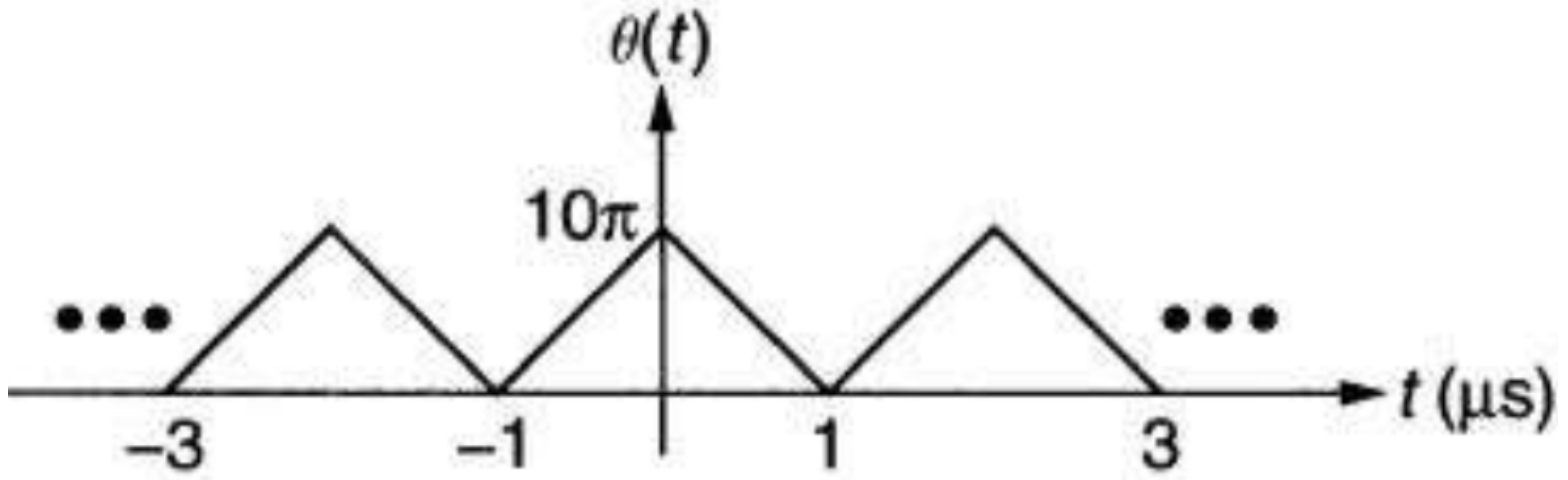


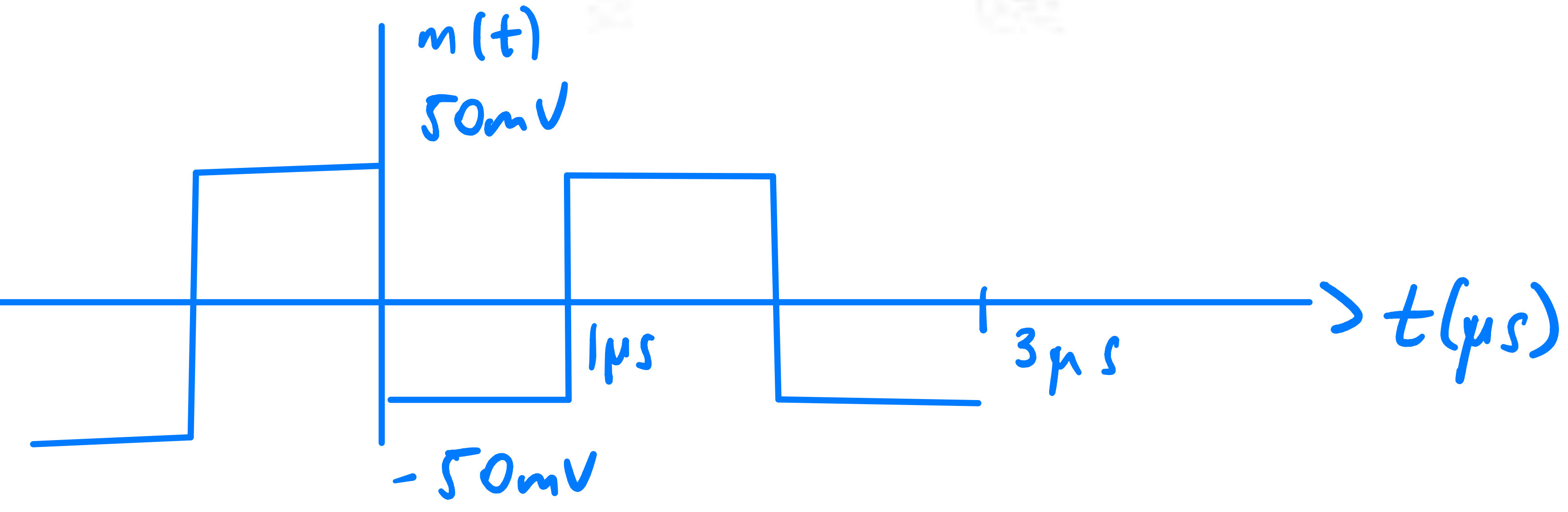
What's the input to my VCO?



$f_c = 1$ GHz

$K_v = 100$ kHz/mV





Find $m(t)$ given $\theta(t)$.

$\theta(t) = 2\pi k_\nu m(t) t$,

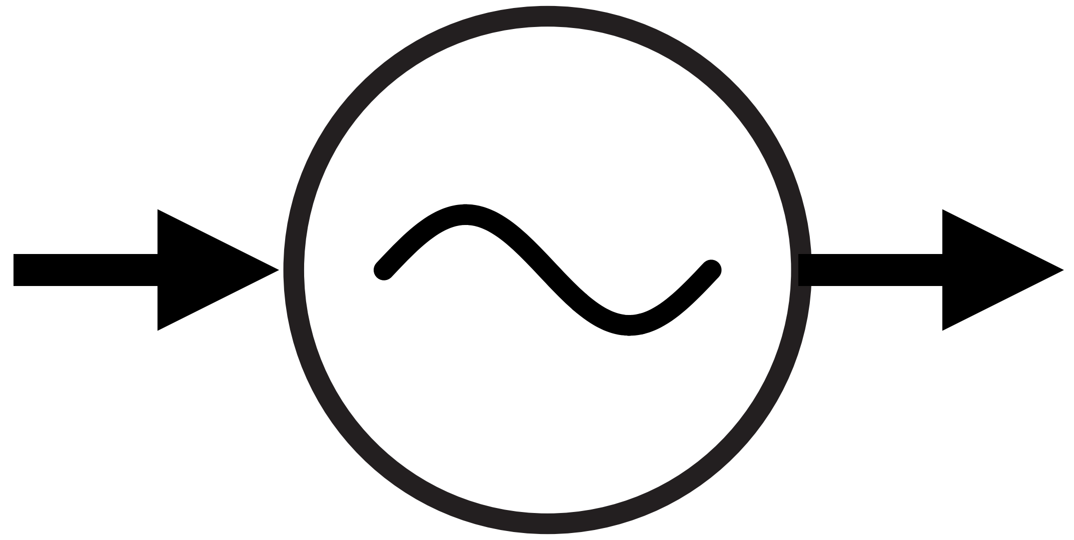
$2\pi k_\nu \max|m(t)| \cdot 1\mu s = 10\pi$

$$\max|m(t)| = \frac{\cancel{10\pi}\ 5}{\cancel{2\pi} \cdot 1\mu s \cdot 100kHz} \ mV = 50mV$$

$$\boxed{m(t) = 50mV \sum_{n=-\infty}^{\infty} (-1)^n I_{[-1/2,1/2]}(t + 1/2 - n)}$$

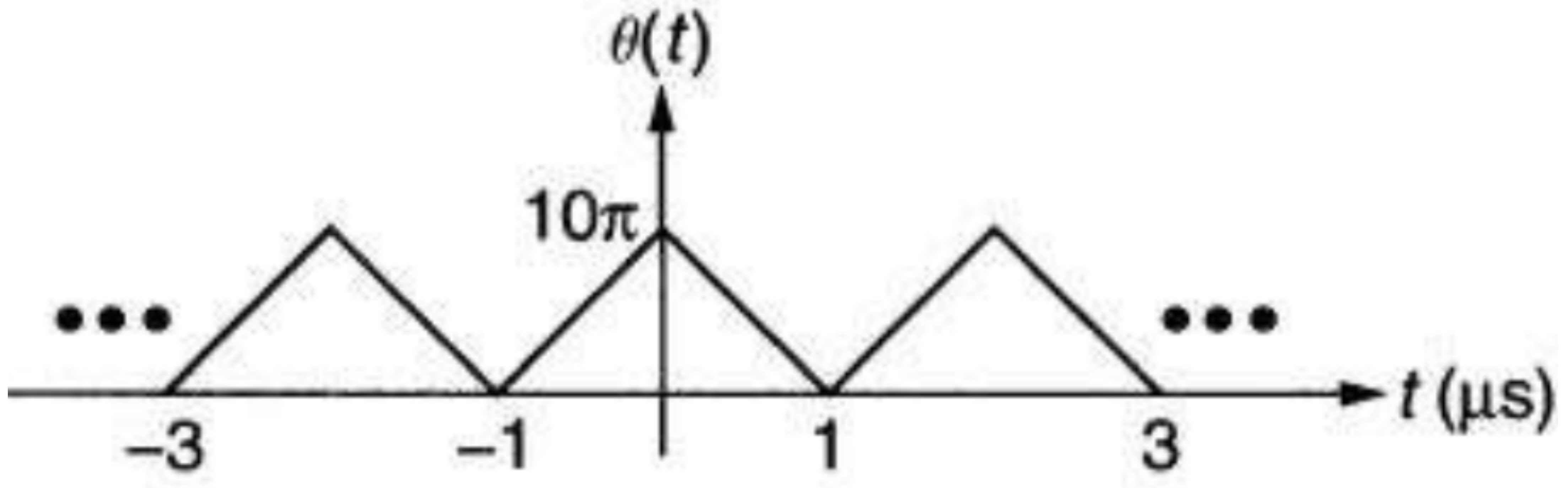
t is in μs

What's the input to my VCO?

$m(t)$  $\cos(2\pi f_c t + \theta(t))$

$f_c = 1$ GHz

$K_v = 100$ kHz/mV



What's the bandwidth of the output?

$B_{FM} = 2B + 2\Delta f_{max}$, $\Delta f_{max} = k_v \max|m(t)| = \frac{100\text{kHz}}{\text{mV}} \cdot 50\text{mV} = 5\text{MHz}$

$B = \frac{1}{2\mu s} = 500\text{kHz}$

$\boxed{B_{FM} = 11\text{MHz}}$